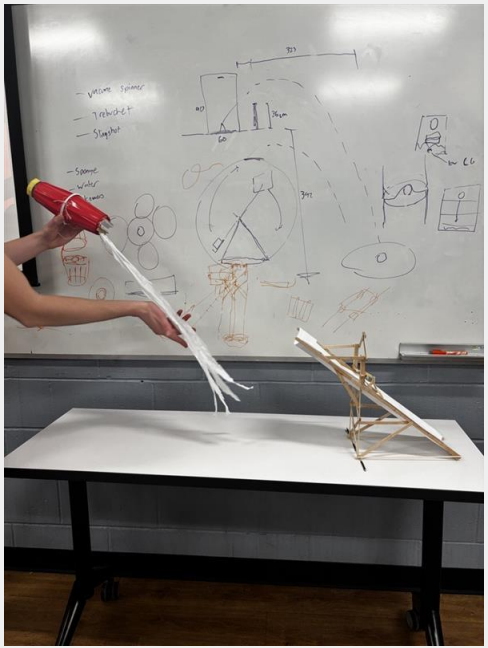


Hermes

Space Launch & Landing Mechanism

Team Jar Jar Binks

Introductions



Matthew Lones
2nd Year
Mathematics & Computer Engineering



Luke Chipman
2nd Year
Mechanical & Materials Engineering



Purujit Kantiya
2nd Year
Mechatronics Engineering



Janet Liu
2nd Year
Mechatronics Engineering

Problem Statement

Current space vehicle **launch and land complexities reduce** the scope of space exploration

Controlled crashing introduces a new way of **increasing** the reach of humanity

Stakeholders



[1]

Canadian Space Agency



Canadian Population



World Governments (Planetary Protection)

Our Challenge

Create a scalable space vehicle launch and land mechanism with low-cost materials, capable of:

1. **Landing** within 30 cm of the target location 3.23 m away from the launch platform
2. **Protecting** an egg from a 3.92 m vertical drop
3. **Clearing** launch vessel minimum 36 cm above launch platform

Constraints

1. **Size -** launch mechanism and vessel must fit within a 60 cm x 60 cm x 90 cm control volume
2. **Activation -** launch mechanism activated with 1 hand
3. **Setup -** launch setup must be completed in 2 minutes

Our Goals

01

Creativity

Design showcases creative thinking

02

Scalability

Scalable design for real world use

03

Feasibility

Practical solution based on physics principles

04

Versatility

Desing can be adapted to fit any situation

Design Process

01

Trade Studies

Research

02A

Calculations & Kinematics

Scientifically Feasible

02B

Initial Design

First Design, V1

03

Design Testing I

Attempt Functional Requirements and Compare Experimental Values

04

Reiteration

Use Testing Results to Inform Design, V2

05

Design Testing II

Repeat Testing on V2 and Calibrate

06

Final Design

Make final changes and document

Mechanism Trade Study

Figures Of Merit	Factor	Vacuum Spinner	Trebuchet	Slingshot	Parachute Landing	Cushioned Landing	Suspended Landing	Sliding Landing
Launch Control	15	0.9	0.8	0.8	0	1	0.8	0
Landing Control	15	0.5	0.8	0.7	0	0.8	1	0.8
Set Up Speed	10	0	0.8	0.7	0.4	0.8	0.8	0
Cost	10	0	0.5	0.8	1	0.5	0.7	0.2
Manufacturability	40	0.3	0.6	0.8	0.9	1	0.9	0
Creativity	5	1	0	0.4	0.5	0.2	0.9	1
Versatility	5	0.2	1	0.5	0.2	1	0.8	0.1
Total Score	100	0.39	0.66	**0.74**	0.535	0.86	**0.865**	0.195

Trade Studies Summary

- Slingshot/Railgun Launch
- Suspended Egg Landing
- Minimize bounce with impact absorption
- Simple and Structurally stable

Feasibility

Manufacturable & Energy Conservation

Scalability

Manufacturable -> Simple

Cost Effective

Simple

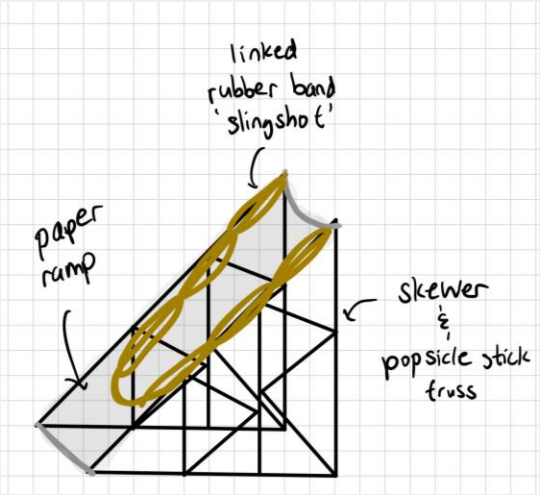
Control

Consistent & Predictable

Initial Design – V1 Details

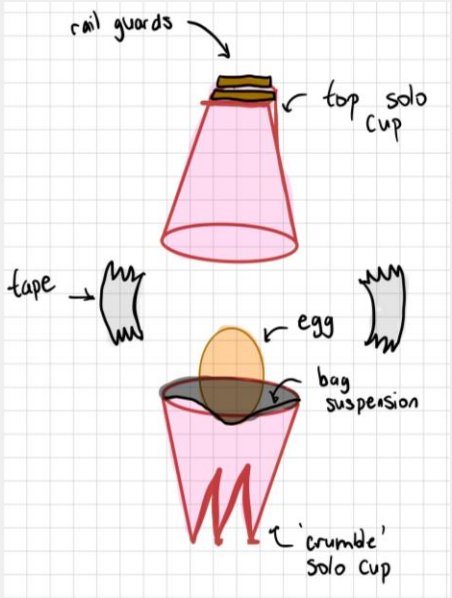
Launch Mechanism:

- Slingshot/Railgun
- Elastic Energy = over rail

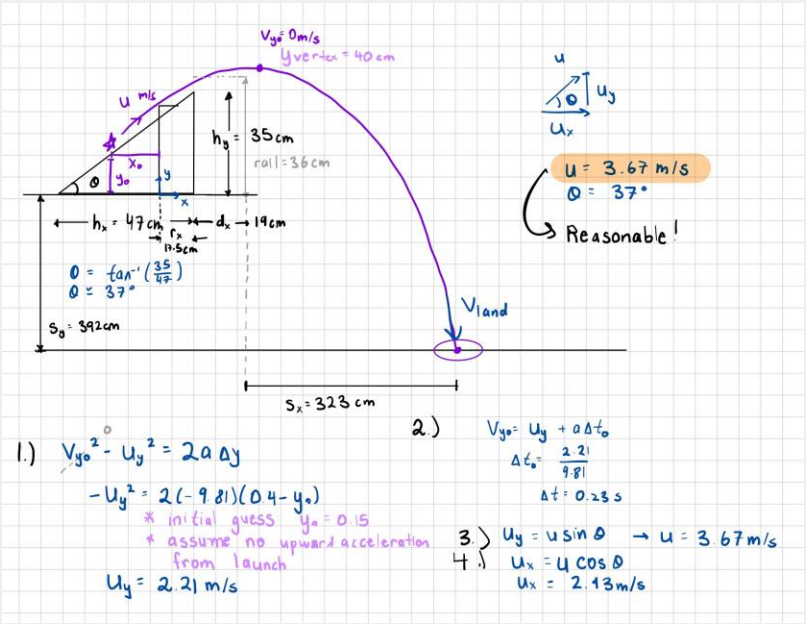


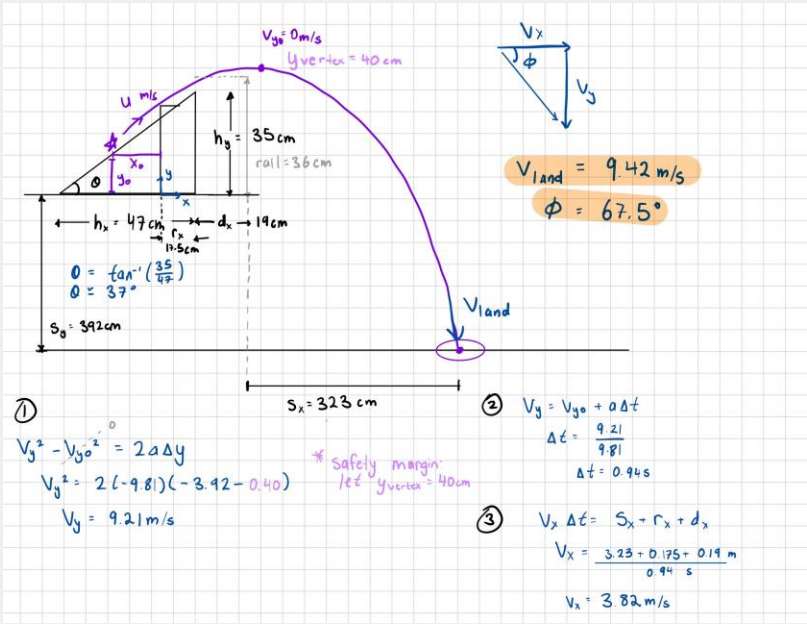
Landing Mechanism:

- Bag Suspends Egg = no break
- Cup crumble = no bounce = accuracy



Initial Design – Kinematics & Feasibility





Planned Two Phase Testing

Iteration 1

🕒 **17:50**

Goal: Determine preliminary design feasability

Iteration 2

🕒 **20:50**

Goal: Optimize launch configuration for accuracy

Test 1 Results + Iteration

01 Structural Strength

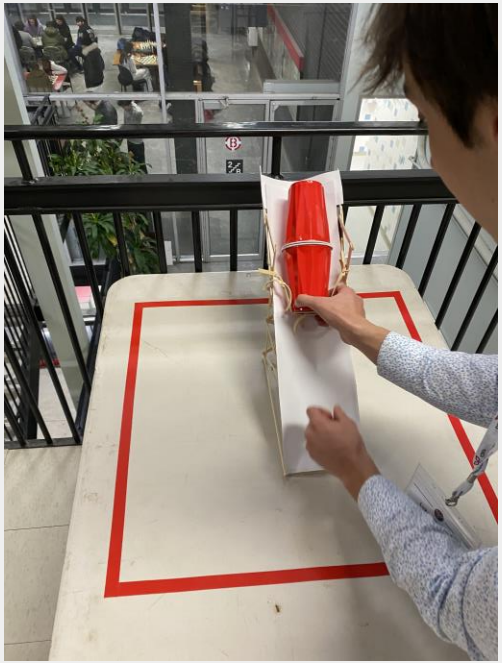
Structural bending observed at elastic band attachment point at highest loading

Fix: Trussing required to withstand loads

02 Landing Shock

Current launch vessel cannot withstand landing impact

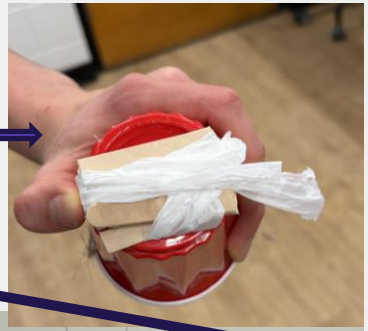
Fix: Slow vessel landing speed and add internal padding

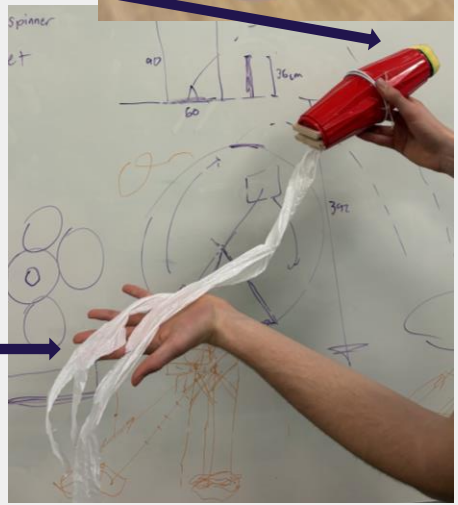


Reiterated Design – V2 Landing

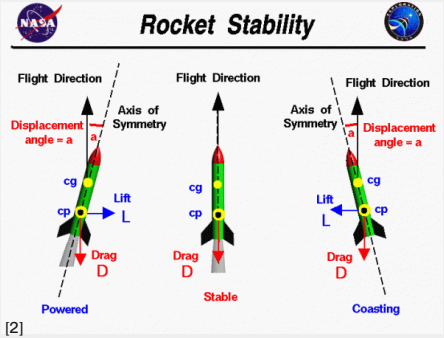
Landing Mechanism V2:

- Furling to reduce launch interference
- Cushioned Bottom
- Restrain Egg and Cushion system to bottom half = lower center of gravity = higher chance of landing on desired face
- Add Streamers = Add Drag and Stability = no bounce









[2]

[2] N. Hall, "Rocket Stability | Glenn Research Center | NASA," *Glenn Research Center | NASA*, Nov. 20, 2023. https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/rocket-stability/

Test 2 Results + Iteration



01 Egg Not Protected

Mostly did not land on sponge

Fix: Resolve Launch/Land inconsistencies

02 Tape and String Durability

Mechanism moved forward, String Unfurled = long prep. time

Fix: Switch tape direction, heat string ends

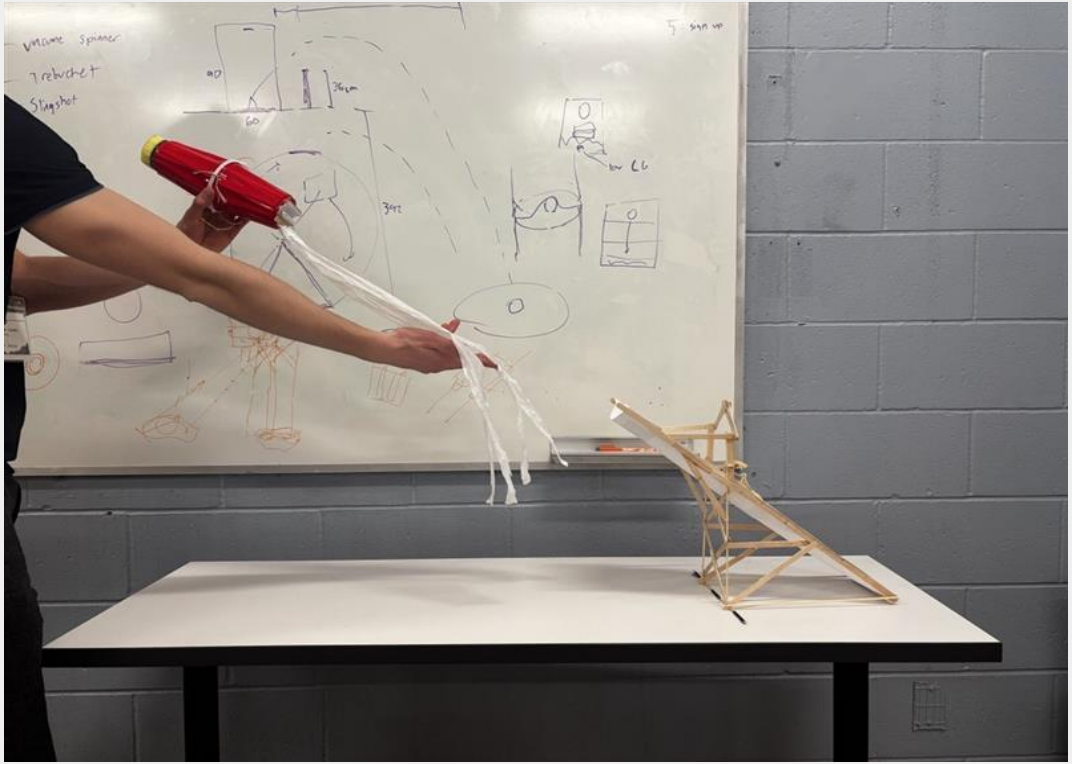


03 Landing/Launch Inconsistency

Current launch is inconsistent. Ranges from 18-20cm tested.

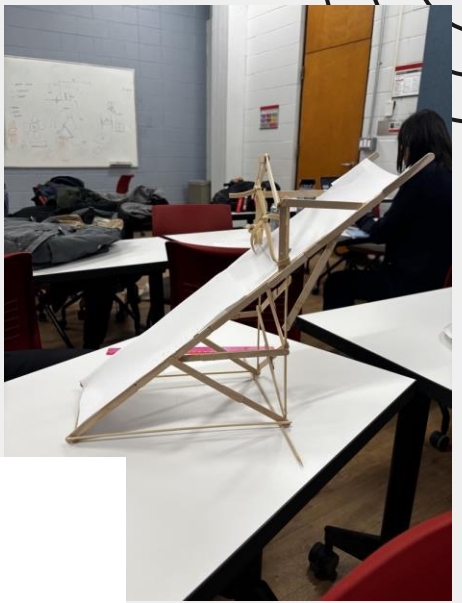
Fix: Slide launch mechanism forward by ~ 10cm and pull back more so vertex is always reached.

Final Design



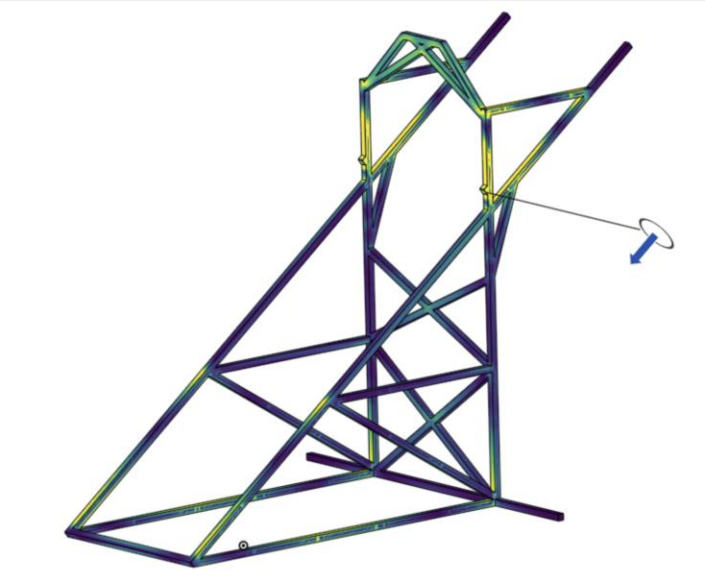
Final Design - Launch Mechanism

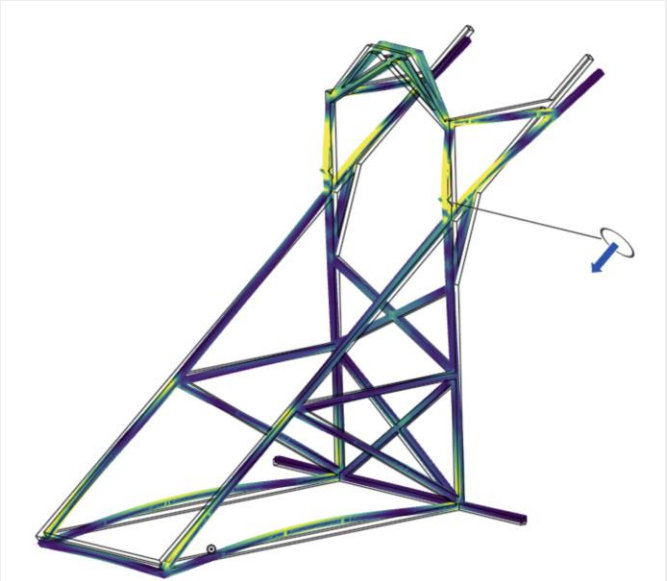
- Trussed structure for high strength
- Elastic thrust mechanism
- Low friction paper launch ramp





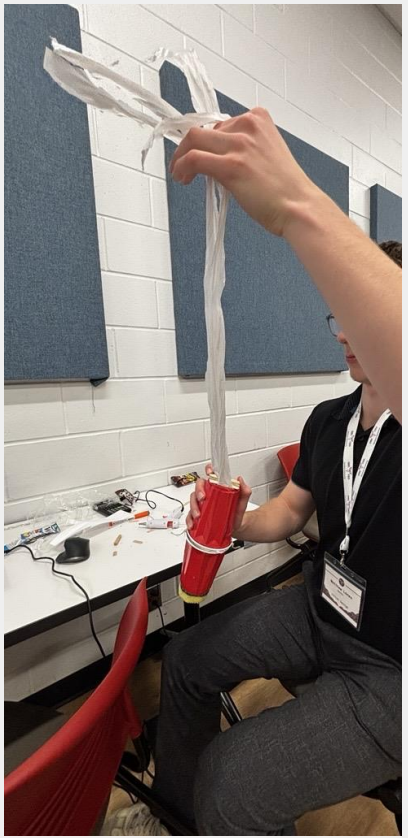
Final Design – Stress Analysis





Launch Vessel

- Streamer for increased drag + lower landing velocity
- Solo cup casing
- Sponge + plastic stuffing for shock absorption
- Popsicle stick mounting mechanism



Bill of Materials

Material	Unit	Cost	Quantity	Total Price
Popsicle Stick	5	100	11	1100
Skewer	1	100	6	600
Hot Glue Stick	1	100	5	500
Elastic Band	2	80	2	160
Garbage Bag	1	400	2	800
Solo Cup	1	250	2	500
Cardstock	1	500	1	500
A4 Paper	1	100	2	200
Twine	30cm	150	2	300
Duct tape	30cm	400	2	800
Test Egg	1	1000	1	1000
Dish Sponge	1	150	1	150
Total				**6610**

Design Feasability

- Simple design that is feasible for real life use
- Similar projects under consideration currently by private companies and NASA [1], [2]





Final Design Evaluation

88%

of predicted performance

Figures Of Merit	Factor	Predicted	Actual
Launch Control	15	0.8	0.8
Landing Control	15	0.7	0.6
Set Up Speed	10	0.7	0.5
Cost	10	0.8	0.5
Manufacturability	40	0.8	0.7
Creativity	5	0.4	0.6
Versatility	5	0.5	0.7
Total Score	100	74	65

Scalability Considerations



Power Source

More power required for heavier launch vehicles



Shock Absorption

Better shock absorption for greater landing impacts



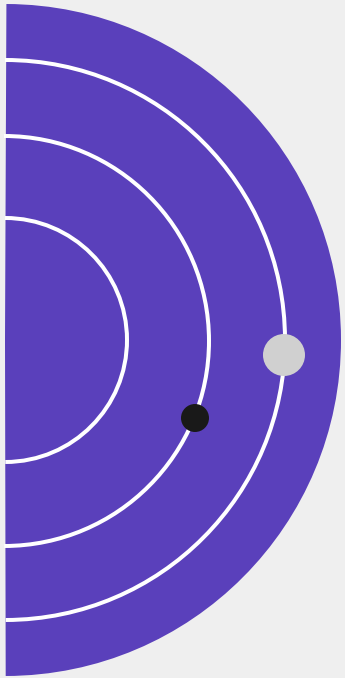
Smoother Streamer Deployment

Controlled and Smooth Streamer Deployment for consistency



Stronger Infrastructure

Better launch mechanism structural stability for High Launch loads



Thank You

References

[1] M. Boucher, “Canadian Space Agency Quietly Releases New Logo,” *SpaceQ*, Mar. 17, 2023. https://spaceq.ca/canadian-space-agency-quietly-releases-new-logo/

[2] N. Hall, “Rocket Stability | Glenn Research Center | NASA,” *Glenn Research Center | NASA*, Nov. 20, 2023. https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/rocket-stability/